Problem Formulation

Define δ_i as :

If $\underline{X} \in$ ***class I*** and ***misclassified*** as ***Class II*** *then* $d(\underline{X})$ *will be* $<0 \Rightarrow \delta_i = -1 \Rightarrow d(\underline{X})\delta_i > 0$

If $\underline{X} \in$ ***class II*** and ***misclassified*** as ***Class I*** *then* $d(\underline{X})$ *will be* $>0 \Rightarrow \delta_i = +1 \Rightarrow d(\underline{X})\delta_i > 0$

Redefine Error function as:

$$\text{Error} = \sum_{\text{all misclassified vectors } (i)} \delta_i \underline{W}^T \underline{X}_i \Rightarrow (\underline{W}^T \underline{X}_i) \text{ takes positive and negative values and } \delta_i \text{ has the same sign}$$

Summation is always ***positive***

Minimize the Error using ***Gradient Descent*** *with step* η

$$\underline{W}^{(updated)} = \underline{W}^{old} - \eta \frac{\partial Error}{\partial W} \quad \textbf{where} \quad \frac{\partial Error}{\partial W} = \sum_{\textit{misclassified} \text{ vectors}} \delta_i \underline{X}_i$$

$$\underline{W}^{(updated)} = \underline{W}^{old} - \eta \sum_{\textit{misclassified} \text{ vectors } \textit{ONLY} \text{ (i)}} \delta_i \underline{X}_i$$